SOLSTICE: TASK 2A

Kailash Anand

too early for next DC batch

§1 Understanding the Situation

NOTE: For any notation related to Bayesian updating, I will re-use what I used in the previous document.

1.1 The Goal of MCMC

- The goal of MCMC is to get samples from some density function, and all other "easy" ways of sampling have failed. This method is especially useful if the density function we want to sample from is weird; in the sense that I can't do inverse transform, or rejection sampling.
- The idea is to create a **Discrete Time Markov Chain** whose:
  1. State space $\mathcal{S}$ is the domain of the density function from which we want to sample (which I will henceforth call the 'Target Density Function': $t(x)$). This could of course also be a PMF.
  2. It's stationary distribution is $t(x)$.
- Once we (somehow) get this Markov Chain, we start doing a random walk. After a certain burn-in period, each state the chain jumps to can be treated as a sample of $t(x)$.
- So now, the question we need an to answer is the following:

$$\boxed{\text{Given a stationary distribution, how can I determine my Markov Chain?}}$$

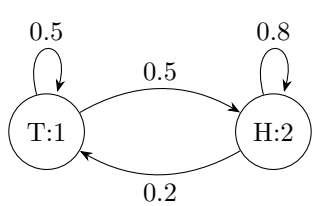
1.2 Finding our Markov Chain

A Markov Chain is determined by its state space $\mathcal{S}$, and its transition probabilities. In our scenario, we have $\mathcal{S} = \text{Domain}[t(x)]$. We need to determine transition probabilities. The only additional information we have to do this is the stationary distribution. To illustrate how we could do this, let's consider an example...

§2 A Coin Toss: Motivation for Detailed Balance

2.1 Some Intuition

An evil scientist cooked up the following crazy coin:[1]



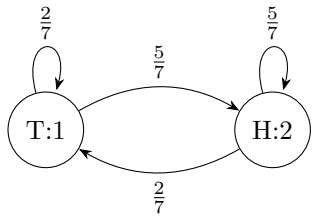
[1] I got these numbers from Prof. John N Tsitsiklis' MIT Lecture on Markov Chains

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}$$

It's limiting distribution is $\pi = \begin{pmatrix} \frac{2}{7} & \frac{5}{7} \end{pmatrix}$

Which means that in the long term, the coin shows up with tails $\frac{2}{7}$ of the time, and heads $\frac{5}{7}$ of time. Hmm...interesting... This tells us two things:

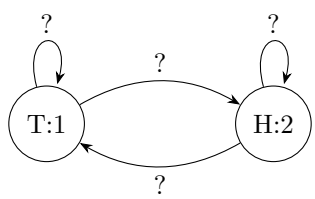
1. This is behaving(in the limit) just like a simple biased coin with $p = \frac{5}{7}$! Which can also be thought of as the following Markov Chain:



2. We can think of the stationary distribution as telling us the long term frequency
3. There seem to be multiple transition matrices that give the same stationary distribution! Let's see this a bit mathematically...

2.2 Uh oh, I have too much freedom...

Consider the following situation - can we determine the transition probabilities given just the limiting distribution π?



$$\pi = \begin{pmatrix} \frac{2}{7} & \frac{5}{7} \end{pmatrix}$$

Say our transition matrix is $P = p_{ij\,(2\times 2)}$
We can write the **Balance Equations:**

$$\pi_1 = \pi_1 \cdot p_{11} + \pi_2 \cdot p_{21}$$

$$\pi_2 = \pi_1 \cdot p_{12} + \pi_2 \cdot p_{22}$$

And the probability constraints:

$$p_{11} + p_{12} = 1$$

$$p_{21} + p_{22} = 1$$

Seems like we have a system of 4 equations and 4 variables...what's the issue then?
Let's set up a matrix equation to analyze the situation:

$$\pi_1 \cdot p_{11} + \pi_2 \cdot p_{21} = \pi_1$$

$$\pi_1 \cdot p_{12} + \pi_2 \cdot p_{22} = \pi_2$$

$$\underbrace{\begin{pmatrix} \pi_1 & 0 & \pi_2 & 0 \\ 0 & \pi_1 & 0 & \pi_2 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}}_{A} \cdot \begin{pmatrix} p_{11} \\ p_{12} \\ p_{21} \\ p_{22} \end{pmatrix} = \begin{pmatrix} \pi_1 \\ \pi_2 \\ 1 \\ 1 \end{pmatrix}$$

The rank of A is 3! (that's an exclamation not a factorial) This means that there is 1 linear dependency between the equations. This means we need to introduce one more constraint. What is a "good" constraint to introduce?

2.3 Some Constraint Options

1. We could fix the value of some p_{ij}. Fixing p_{11} to 0.5 gives us the 'Evil Scientist Markov Chain', fixing it to $\frac{2}{7}$ gives us the simple biased coin markov chain.
2. We could introduce another 5th equation in p_{ij}'s which is linearly independent of the previous 4.

These methods seem simple, but they have some issues:

- How do you find another equation which is linearly independent, while also ensuring that ALL the p_{ij}'s are positive?
- Again a similar issue arises, how do you guess a p_{ij} such that all the other $p_{ij} \geq 0$?

Alright, we tried thinking about a solution to this by staring at algebra. Let's take a step back and look at a different perspective.

2.4 I've done it right? Where's the Applause? - Almost Detailed Balance

Let's go back to the coin toss example. In the case of the 'Evil Scientist Markov Chain' coin toss, using the transition matrix, we were able to find the stationary distribution. Then we realized, that this could just as well be a simple bias coin. Interesting. Can we extend that to more states? Well yes of course, consider a faced dice with some crazy transition probabilities, but its limiting distribution is

$$\pi = \begin{pmatrix} \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \end{pmatrix}$$

This is just a biased die, with probability of landing on 1 being $\frac{1}{4}$, landing on 2 being $\frac{1}{3}$, and so on.

This is a HUGE perspective shift! We have replaced a very complicated dice with very weird short-term behavior, with a simple dice which always behaves the same way. What would the markov chain of such a die look like?

Well, it would look like this -

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{12} & \frac{1}{18} & \frac{1}{9} \end{pmatrix}$$

No matter where I start, every output state will be π! Yay! We have solved the problem!!...Right? RIGHT? Where's the applause?

Where's the applause?

This method helps us sample from PMFs. But that wasn't hard to do in the first place. We want to sample from a PDF whose density we know upto only a proportionality factor. We now reach our ultimate constraint, beyond matrix algebra, linearly dependencies and guesswork. We need to extend this to PDFs, keeping in mind that we don't know the normalizing constants.

A good place to start is with ratios. Why?

1. Say we knew $t_1(x) \propto t(x)$, where $t_1(x)$ has the same shape but without the normalizing constant. Then $\dfrac{t_1(x)}{t_1(y)} = \dfrac{t(x)}{t(y)}$
2. Ratios of densities behave like ratios of discrete probability masses. This means this idea can be used for both PMFs and PDFs.

$$\frac{\mathbb{P}(X = a)}{\mathbb{P}(X = b)} = \frac{f_X(a) \cdot \delta}{f_X(b) \cdot \delta} = \frac{f_X(a)}{f_X(b)}$$

3. Ratios let us think *locally*, in a pairwise fashion, instead of *globally*. Global constraints aren't strong enough, we have 1 extra linear dependency, which is hard to fix. Also worth noting, is that this is good when extending our state space $\mathcal{S}$ to be $\mathbb{R}$.

2.5 Let's turn to Ratios: Detailed Balance

Okay, now we have a new philosophy. We are no longer trying to do linear algebra and solve matrix equations, we are here to find some **local constraint**, which will be in the form of **ratios**, that ensures that the stationary distribution is exactly the distribution we want to sample from.

Let's explicity see the equations for 3 state Markov Chain, and look for some local ratio structure that could work.

$$\begin{pmatrix} \pi_1 & \pi_2 & \pi_3 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} = \begin{pmatrix} \pi_1 & \pi_2 & \pi_3 \end{pmatrix}$$

We tried just making all the $p_{ij} = \pi_j$ earlier, but that wasn't useful. However it gave a valid solution.

The idea behind that is, if all the $p_{ij} = \pi_j$'s are equal, we could just pull that out, the remaining terms will sum up to 1, leaving us with that π_j. Can we find a similar trick, but with local pairwise ratios as constraints?

Consider:

$$\pi_1 \cdot p_{11} + \pi_2 \cdot p_{21} + \pi_3 \cdot p_{31} = \pi_1$$

We earlier used the fact that $\pi_1 + \pi_2 + \pi_3 = 1$, and made all the $p_{ij} = \pi_j$. Let's try to use $p_{11} + p_{12} + p_{13} = 1$.

But there is no p_{12} in the balance equation, let's see if setting $p_{12} = p_{21}$, i.e. making the martrix symmetric, works.

$$\pi_1 \cdot p_{11} + \pi_2 \cdot p_{12} + \pi_3 \cdot p_{13} = \pi_1$$

Well, this kind of does nothing unless $\pi_1 = \pi_2 = \pi_3$. Then we could pull that out as a common factor and we'd be done.

What about setting $\pi_2 \cdot p_{21} = \pi_1 \cdot p_{12}$?

This works! We swap not only the the subscripts, but also the π_i that we multiply with!

This leads us to the crown-jewel of MCMC:

Detailed Balance: $\boxed{\pi_i \cdot p_{ij} = \pi_j \cdot p_{ji}}$

This satisfies all our requirements:

1. We don't need π_i or π_j exactly, knowing them upto a proportionality factor works because...
2. We are working with ratios $\rightarrow \dfrac{p_{ij}}{p_{ji}} = \dfrac{\pi_j}{\pi_i}$ This also extends nicely to the continuous domain.
3. These are local pairwise constraints. It intuitively "feels" like a flow conservation equation.

Some important points:

1. We have not put number on anything yet. We have only figured a constraint, that when satisfied *guarantees* the stationary distribution to be the PDF/PMF we want to sample from. This leads us to the next sections: algorithms to determine the transition probabilities.

2. Detailed Balance is a sufficient condition to ensure that the stationary distribution is our required PDF/PMF. $\left(\pi_i \cdot p_{ij} = \pi_j \cdot p_{ji} \implies \underline{\sum_i \pi_i \cdot p_{ij}} = \sum_i \pi_j \cdot p_{ji} = \underline{\pi_j}\right)$ However, it is not a necessary condition.

§3 Metropolis-Hastings Algorithm

Idea: Start with an initial guess for the markov chain, then using these "proposal" probabilities, calculate transition probabilities p_{ij} such that *detailed balance* is satisfied.

I implemented the Metropolis-Hastings Algorithm for a dice with different proposals. Please find the notebook in the Notebooks section.

3.0.1 Notation

- $g(i,j)$: The probability of proposing state j given we are at state i
- $A(i,j)$: The probability of accpeting state j given we are at state i.
- $\delta[n]$: The discrete delta function. It takes a value of 1 when $n = 0$, and is 0 everywhere else. It can also be thought of a difference in step functions which will be useful when trying to extend these ideas to continuous spaces.
- p_{ij} or $P(i,j)$: The transition probability of going to state j given we are at state i.

My Learnings:

1. To find the transition matrix, self-transition includes not only being proposed to stay in the same state by `g(i,j)`, but also rejecting moving to other states.
   So, $p_{ii} = g(i,i)\underbrace{A(i,i)}_{=1} + \sum_{k\neq i} g(i,k)(1-A(i,k)) = g(i,i) + \sum_{k\neq i} g(i,k) - \sum_{k\neq i} g(i,k)A(i,k) = \boxed{1 - \sum_{k\neq i} g(i,k)A(i,k)}$

2. A general formula for transition probabilities:

$$p_{ij} = g(i,j)A(i,j) + \delta(i-j)\sum_{k\neq i} g(i,k)(1-A(i,k))$$

   Since $A(i,i) = 1 \forall i \in \mathcal{S}$,

$$p_{ij} = g(i,j)A(i,j) + \delta(i-j)\sum_{k} g(i,k)(1-A(i,k))$$

$$p_{ij} = g(i,j)A(i,j) + \delta(i-j)\left(\underbrace{\left\{\sum_k g(i,k)\right\}}_{=1} - \underbrace{\left\{\sum_k g(i,k)A(i,k))\right\}}_{\text{Define this to be } Q(i)+g(i,i)}\right)$$

   $Q(i)$ is the probability of leaving state i in the markov chain.

General formula: $\rightarrow \boxed{p_{ij} = g(i,j)A(i,j) + \delta[i-j]\left(1 - Q(i) - g(i,i)\right)}$

3. If I make a state "sticky"(in the code I chose state 3), then though the proposal probability is high, it gets rejected by `A(i,j)` so much, and if proposed to go away from 3, the acceptance is 1. So until I let the simulation run past 5000 or so loops, the frequency of hitting state 3 is very low.

4. The samples are correlated. That's good because it means we stay in a region of high density for a longer time. However if the autocorrelation is too high, then we might not be exploring the tails of the function which is bad.

5. There is also a commented part where I try to sample from a PDF by discretizing the real axis into 'buckets'. This helped me understand how one would go from a discrete state space to a continuous state-space.

3.1 Continuous State Space Metropolis

3.1.1 Notation

- $g(j|i)$: The proposal probability density function
- $A(i,j)$: The probability of accpeting state j given we are at state i. This is NOT a density function, it is finite number.
- $\delta(x)$: The dirac delta distribution. It is the derivative of the unit step function.
- $p(j|i)$: The transition probability density function

I think of a continuous state space MCMC as having a fixed interval, discretizing into n slots, then letting $n \to \infty$.

1. As $n \to \infty$, `len(PI)` $\to \infty$, but the value of element:

   $$\texttt{PI[i] = t_CDF(x_coords[i+1]) - t_CDF(x_coords[i])} = \left(\frac{\texttt{t_CDF(x_coords[i+1]) - t_CDF(x_coords[i])}}{\delta}\right)\cdot\delta$$

   $$\boxed{\texttt{PI[i]} \approx \pi(x_i)\cdot\delta_1}$$

   δ_1 is width of each interval. This reminds us a lot of how we went from PMFs to PDFs, telling us that our stationary distribution π should be now thought of as a PDF.

2. The proposal function `g(i,j)` has an inverse relation to `len(PI)`, meaning that this also becomes a PDF in the limit, with the SAME factor with which `PI` became infinity. So this can also be safely replaced with something like: `g(i,j)` $\approx g(X_{n+1} = j|X_n = i)\cdot\delta_1$. I'll represent this density function with the shorthand: $g(j|i)$. This conditional density g should be one we can easily sample from. So a uniform or a normal distribution is a good choice.

3. The acceptance probability is still a number, it doesn't become an infinitesimal quantity. When we write $A(i,j)$ it will have some δ_1 factors, if we cancel all of them out we get:

   $$A(i,j) = \min\left\{1, \frac{g(j|i)}{g(i|j)}\cdot\frac{\pi(x_j)}{\pi(x_i)}\right\}$$

4. The transition matrix probabilities will also be density functions now. Let us extend the discrete version of the formula to the continuous domain.

   In the discrete world:

   $$p_{ij} = g(i,j)A(i,j) + \delta[i-j]\left(1 - Q(i) - g(i,i)\right)$$

$$\boxed{\begin{aligned}
&p_{ij} \approx p(j|i)\delta_1 \\
&g(i,j) \approx g(j|i)\delta_1 \\
&\delta[i-j] \approx \delta(i-j)\delta_1 \\
&Q(i) \approx \sum_{k \neq i} \left(g(k|i)\delta_1\right) A(i,k) = \int_{\mathcal{S}} g(k|i)A(i,k)\ dk
\end{aligned}}$$

Therefore in the continuous world,

$$p(j|i)\delta_1 = g(j|i)\delta_1 A(i,j) + \delta(i-j)\delta_1 \left(1 - Q(i) - g(i|i)\delta_1\right)$$

Cancelling the δ_1's, we get:

$$p(j|i) = g(j|i)A(i,j) + \delta(i-j)\left(1 - Q(i) - g(i|i)\delta_1\right)$$

Since $\delta_1 \to 0$ and $g(i|i)$ is finite, $g(i|i)\delta_1 \to 0$. Basically, $g(i,i)$ represents probability mass at a single discrete point, and in the limit that the space becomes continuous, mass at a single point vanishes.

$$\boxed{p(j|i) = g(j|i)A(i,j) + \delta(i-j)\left(1 - Q(i)\right)}$$ [2]

This shows us that unlike in a finite state markov chain where we have transition *probabilities*, in a continuous state space we have analogous transition *densities*! In a finite state markov chain, we split the line from 0 to 1 into $n = n(\mathcal{S})$ intervals, and use the CDF of the PMF to figure out the proposal state. As we let $n \to \infty$, this way of choosing a proposal state becomes equivalent to sampling using inverse CDF from the *transition density function*.

5. The issue of figuring out a good proposal function is fixed if we just make it symmetric. Then the two factors will cancel out. This is sort of the same trade-off we had with finding a good prior in task-1. The biased prior (sticky proposal) is good if π is also like that, because the burn-in time will be much shorter. However if it isn't as discussed in the earlier section, it will backfire.

3.2 Using Metropolis-Hastings for a PDF

I have done this in-detail in a python notebook which can be found in the Notebooks section.

3.3 Limitations

1. If the target distribution is spiky, then we need to run many loops to get to steady state. Further, the burn-in time is heavily dependent on our initialization point.

2. The proposal distribution $g(i|j)$ plays an important role in deciding the mixing time, so we need to optimize for that as well

[2] $\delta[n]$, with square brackets is the discrete delta function. $\delta(x)$ without the square brackets is the dirac delta distribution. δ_1 is a number and is the width of each interval while discretizing the state space.

§4 Gibbs Sampler

This method is used to sample points from a high dimensional distribution whose marginals are easy to sample from. It is a special case of the metropolis algorithm where the transition probabilities are determined by the marginals. It also satisfies detailed balance, and thus results in the correct stationary distribution in theory.

I've implemented the Gibbs Sampler in a python notebook which can be found in the Notebooks section.

Learnings:

1. It's acceptance rate is 1, so you get many more samples much quicker. Further, it is much easier to implement in code, without having to define proposals, and acceptance probability functions.
2. However a huge constraint is that you must be able to sample from the conditional distributions.
3. The autocorrelation between successive samples is much lower than MH, it is nearly i.i.d. However as the random variables themselves get more correlated, the autocorrelation between samples also increases to a large number.
4. This seems to be a general issue with MCMC, which is that if the target distribution is 'spiky', then you either will never reach the tails, or you might never reach the spike. If there are multiple high density areas then the result will be even worse. In general, the variance shouldn't be 'very' large. This sort of ties to the assumption of *irreducibility* we made at the start. By getting high spikes in the density, we are sort of creating multiple recurrence classes, meaning we could get "stuck" in a region, kind of like if the markov chain was reducible.

§5 Monte Carlo Integration

Also in one of the notebooks, which can be found in the Notebooks section.

§6 Notebooks

Click Here for the folder with all the notebooks

1. MCMC: Contains PMF MCMC, and the limiting case of discrete space MCMC becoming continuous space MCMC. That code is commented out by default. Uncomment that code, and you also need to re-center the x-coordinates.

2. Continuous Space MCMC: I used Metropolis-Hastings to sample from a "weird" target density function. Then I used Monte-Carlo Integration to find it's area to re-normalize it.

3. Gibbs Sampling: Gibbs Sampling code to sample from bivariate normal distribution